Tasmin Khan

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Education

Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering (EEE)
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

- CGPA: **3.80 / 4.00** — Feb'2020 - Mar'2025
- **Dean's List Award**, BUET (Awarded for Average GPA $\geq$ 3.75 in two consecutive terms) — 2023, 2021
- **University Merit Scholarship**, BUET (Awarded based on merit ranking in term examinations) — 2021

Relevant Coursework

- VLSI Circuits and Design [Lab]
- Analog Integrated Circuits & Design [Lab]
- Digital Electronics [Lab]
- Microprocessor & Embedded Systems [Lab]

VLSI Competition

Champion in the VLSI-thon arranged by Neural Semiconductor — Dec'2024

- Identified the error source in a circuit from the output clock pattern and designed the solution logic circuit
- Designed the circuit and layout to implement logic from a given truth table, along with DRC and LVS checks

Research Experience

Hardware Architecture & RTL Automation, (Apr'2026-Present)
Principal Investigator: Dr. Rakin Haider, Assistant professor, Computer Science and Engineering (CSE), BUET
Funded by: Bangladesh Industry Research, Development and Innovation (BIRDI)

- Leading design development: authoring design specifications, delegating tasks, and designing RTL with testbenches including AMBA bus interconnect, memory controller and systolic array for GEMM operation
- Exploring sparsity-enabling architectures for inference hardware
- Led training sessions teaching the research team SystemVerilog, hardware design, and testbench methodology
- Investigating limitations of existing datasets (VerilogDB, VeriCoder, RealBench) for industrial-scale designs
- Testing staged spec-to-RTL generation via intermediate representations to improve LLM design performance

Search Towards an Advanced Deep Learning Algorithm for Magnetic Resonance Imaging (MRI) Quantitative Susceptibility Mapping (QSM) with Magnetic Anisotropy (UG Thesis) [GitHub]
Supervisor: Dr. Maruf Ahmed, Assistant Professor, EEE, BUET

- Developed 2D & 3D datasets by implementing a complete multi-orientation MRI preprocessing pipeline
- Implemented CNN and Transformer models with PyTorch, finding CNNs more suitable due to localized anisotropy

Development of Low-Cost Methods for Monitoring and Early Warning for Geo-hazards [GitHub, Live]
Principal Investigator: Dr. Tahmeed Malik Al-Hussaini, Professor, Civil Engineering (CE), BUET
Co-Principal Investigator: Dr. Md Zunaid Baten, Associate Professor, EEE, BUET
Funded by: Research and Innovation Center for Science and Engineering (RISE) Internal Research Grant, BUET

- Conducted noise analysis of MEMS sensors, determining the optimal device for signals down to 0.1 m/s^2
- Built stand-alone unit, validated in lab, and field-deployed a multi-station network detecting multiple earthquakes

Industrial Experience

Modeling Engineer — **Apr'2026 - Present**
gixsystems-AI, Dhaka, Bangladesh

- Behavioral Modeling and Design of Power Converter Integrated circuit and Discrete Devices
- Built a physics-informed DL pipeline mapping characteristic curves to compact model parameters
- Established open-source pipeline with ngspice and OpenVAF for simulation using custom process design kit (PDK)

Internship on VLSI Design — Jun - Jul'2024
Silliconova, Dhaka, Bangladesh

- Study of Digital Design and Verification, Physical Design, DFT, Packaging and FinFET, GAA, CFET technologies
- Hands-on experience in standard cell Layout Design, DRC/LVS Verification, and Placement Techniques

Publications

Under Review in IEEE Transactions of Instrumentation and Measurement May'2026
"Design and Development of Low-Cost Earthquake Monitoring System Based on MEMS Accelerometer"
Tasmin Khan, Rahat Ibn Nabi, Nilotpaul Kundu Dhrubo et al

Presented at IAGA/IASPEI Joint Scientific Meeting | Lisbon, Portugal Aug - Sept'2025
"Development of Low-Cost Seismic Monitoring System for Field Application in Bangladesh"
Al-Hussaini, T.M., Baten, M.Z., Nabi, R.I., Islam, K., Ghosh, P., **Khan, T.** et al

Projects

- **AXI4-interconnect with weighted arbiter for fair SoC bandwidth distribution** [GitHub]
Designed the crossbar with custom arbiter weight for fairness per transaction
- **Three Band Digital Audio Equalizer Peripheral with biquad IIR filters** [GitHub]
Designed three-band audio equalizer with 16-bit fixed-point biquad IIR filters compiled using Icarus Verilog and verified with cocotb Python testbenches, implementing a RISC-V peripheral for digital audio processing
- **16bit Ripple Carry Adder with Coverage Driven Layered Testbench and Layout** [GitHub]
Developed a layered testbench in NCSim/Xcelium skewing randomization toward unhit coverbins, and performed PnR in Innovus, achieving faster coverage closure and a 93.71% density-utilized layout
- **Hardware Gaming Interface to play Single Round Tic-Tac-Toe with Custom PCB** [GitHub]
Built game logic with player identification and cheat prevention designing PCBs in Proteus for an error-free play

Teaching Experience

Adjunct Lecturer Dec'2025 - Apr'2026
Department of Electrical and Electronic Engineering (EEE)
Ahsanullah University Of Science and Technology (AUST), Dhaka, Bangladesh

Physics & ICT Instructor 2021-2023
Udvash Academic Coaching, Grade: 8 - 12

Technical Skills

Design & Implementation: Cadence (Virtuoso, Genus, Innovus, Assura), Vivado, Quartus, OrCAD PSpice, Proteus
Simulation & Verification: Xcelium, ModelSim, QuestaSim, Icarus Verilog, Aldec Riviera-PRO, Cocotb, Pytest
Programming & Modeling: Verilog HDL, SystemVerilog, Python, C, C++, PyTorch, Git, MATLAB, Simulink
Development Platforms: Linux, Keil uVision, Makefile, Arduino IDE, FPGA and CPLD Development Boards
Visualization & Documentation: GTKWave, SimVision, Matplotlib, LaTeX, AutoCAD, Adobe Illustrator

Awards

- **RISE Undergraduate Student Research Grant** of ∼700 USD for Thesis 2024
- **Honorable mention** in Intra BUET PCB Design Contest arranged by
IEEE Robotics and Automation society (RAS), BUET Student Branch Chapter 2023
- **Second runner-up** in 'HULT Prize at BUET' - A Social Entrepreneurship Competition 2020
- **Government Merit Scholarship**, Ministry of Education, Bangladesh 2019, 2017

Standardized Test

IELTS Academic: Overall 8.5 | Listening: 9.0 | Reading: 9.0 | Writing: 8.0 | Speaking: 8.0 Aug'2025

Leadership Activities

- Chairperson, IEEE WIE Affinity Group, BUET Student Branch 2024
- Assistant General Secretary, IEEE WIE Affinity Group, BUET Student Branch 2024
- Executive Member, IEEE Bangladesh Section, BUET Student Branch 2021-22